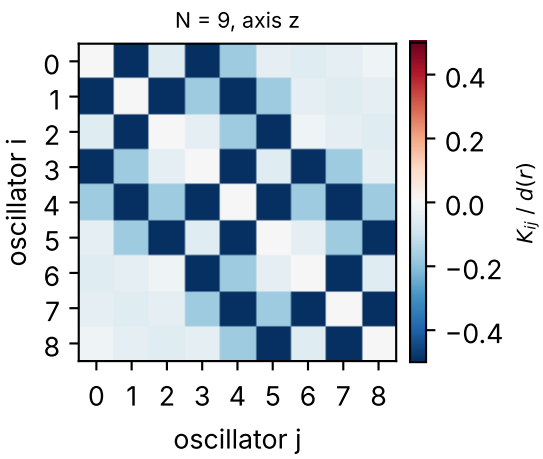
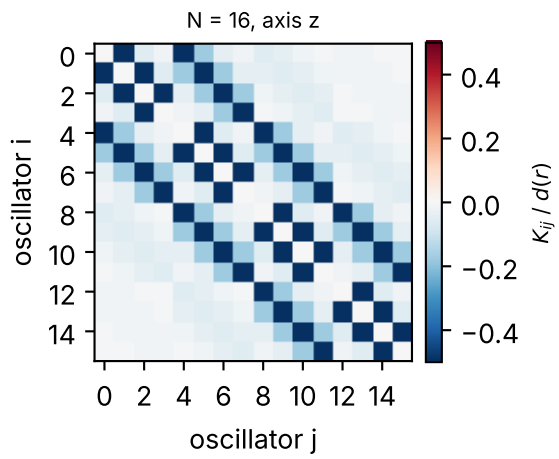
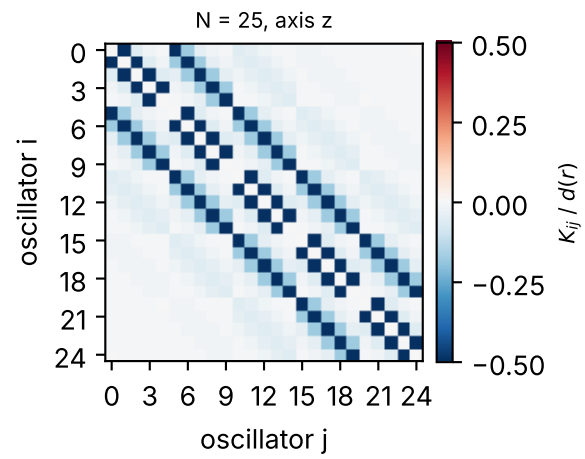
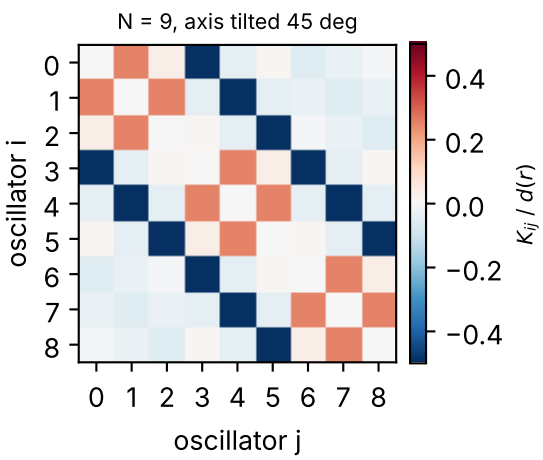
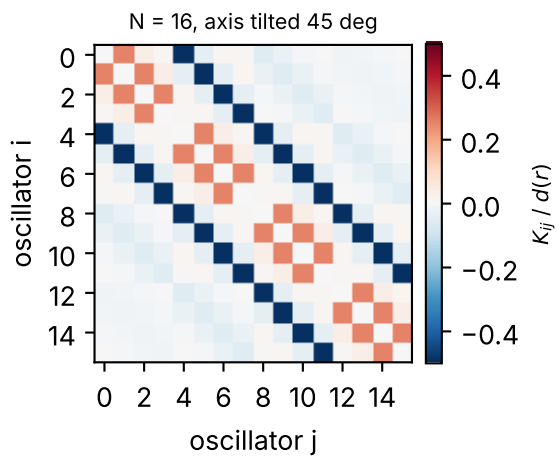
Geometry-defined dipolar coupling graphs (spacing r = 80 nm, dipolar threshold 5 % of max |K|)
PRELIMINARY - analytic first-order K_ij, not yet checked against macrospin runs; no training epochs/seeds involved

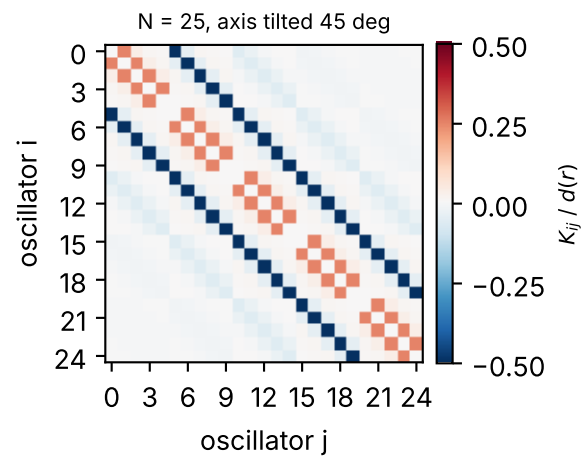


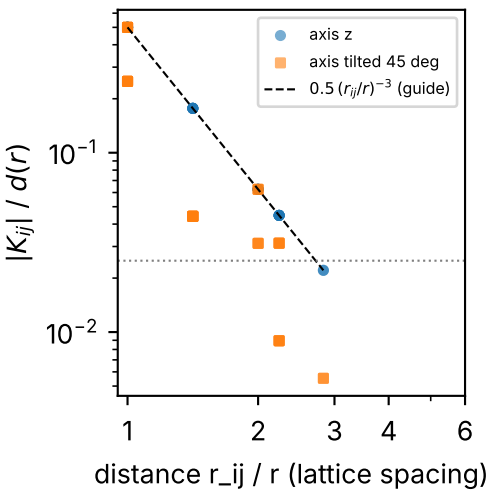


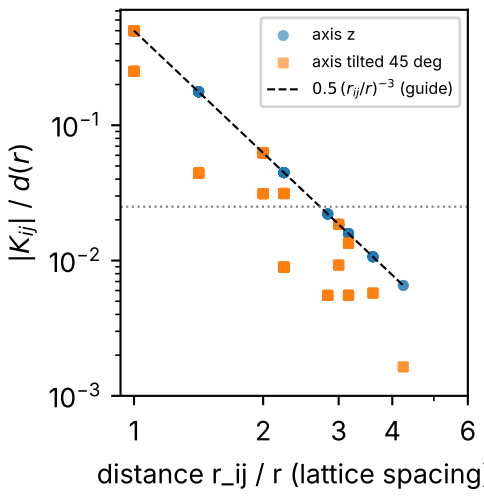


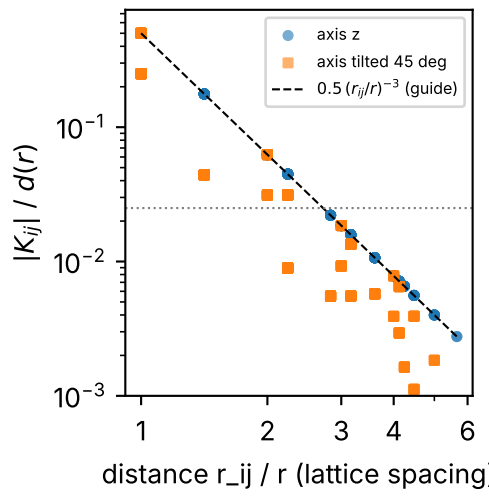












N	graph	edges	mean degree	min degree	components	diameter (hops)
9	NN square	12	2.67	2	1	4.0
9	4NSQ (<=2r)	26	5.78	5	1	2.0
9	dipolar axis z (>5%)	34	7.56	7	1	2.0
9	dipolar axis tilted 45 deg (>5%)	30	6.67	6	1	2.0
9	all-to-all	36	8.0	8	1	1.0
16	NN square	24	3.0	2	1	6.0
16	4NSQ (<=2r)	58	7.25	5	1	3.0
16	dipolar axis z (>5%)	82	10.25	7	1	2.0
16	dipolar axis tilted 45 deg (>5%)	70	8.75	6	1	3.0
16	all-to-all	120	15.0	15	1	1.0
25	NN square	40	3.2	2	1	8.0
25	4NSQ (<=2r)	102	8.16	5	1	4.0
25	dipolar axis z (>5%)	150	12.0	7	1	3.0
25	dipolar axis tilted 45 deg (>5%)	126	10.08	6	1	3.0
25	all-to-all	300	24.0	24	1	1.0